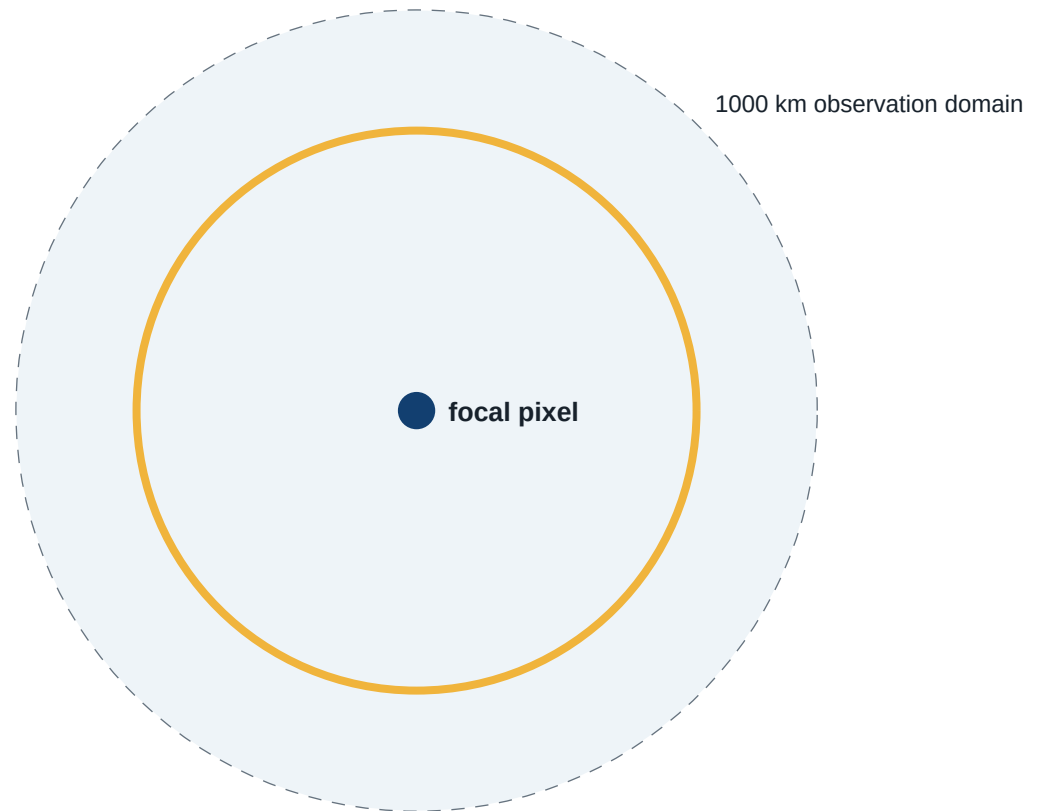


# 1. Existing fixed-radius method

The validated baseline stays intact.

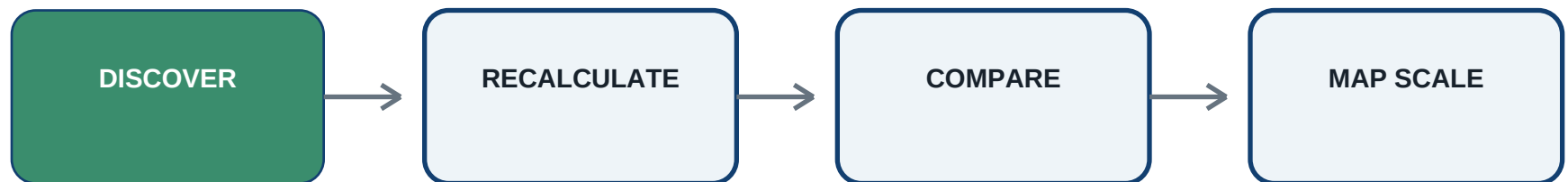
1. Each focal pixel uses one imposed distance.
2. Cold is lower-tail TMIN; warm is strict upper-tail TMAX.
3. Primary Delta is the median of pairwise cold-minus-warm values.



## 2. Why one universal distance may be limiting

A fixed domain can mix different geographic regimes.

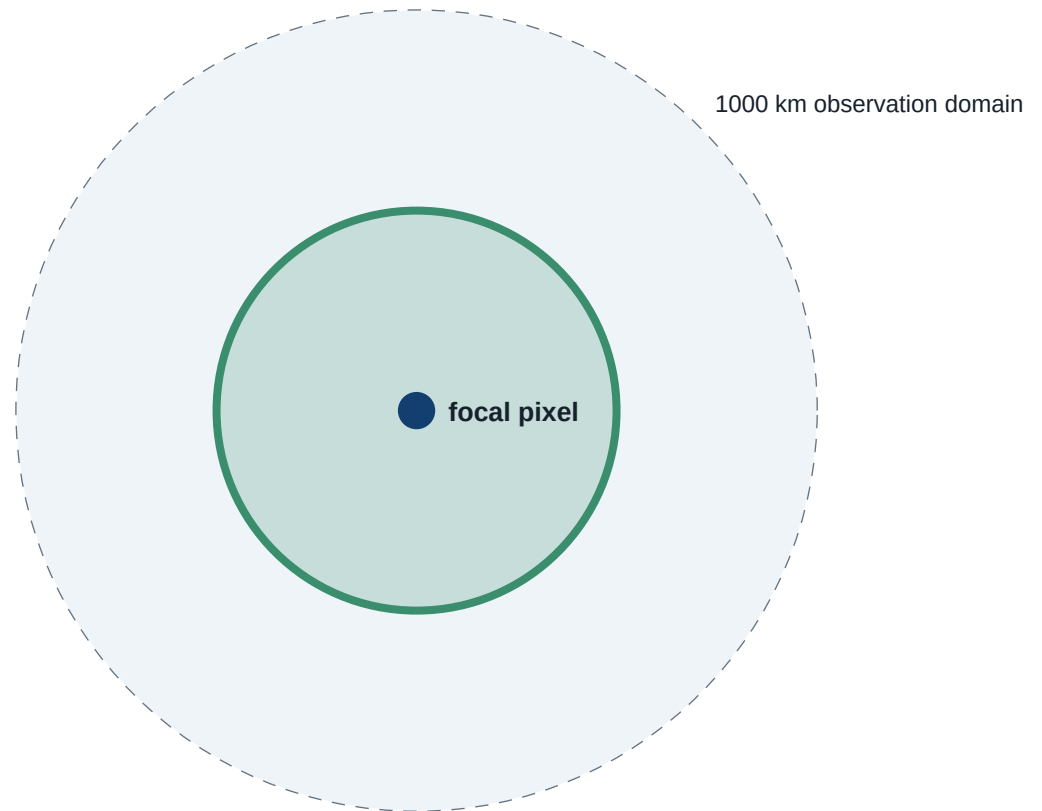
1. A short local regime can be diluted by far-field relationships.
2. A broad regime can be truncated by a small radius.
3. The experiment tests variation; it does not assume adaptation is better.



### 3. One focal pixel

Start with one location and preserve pair identity.

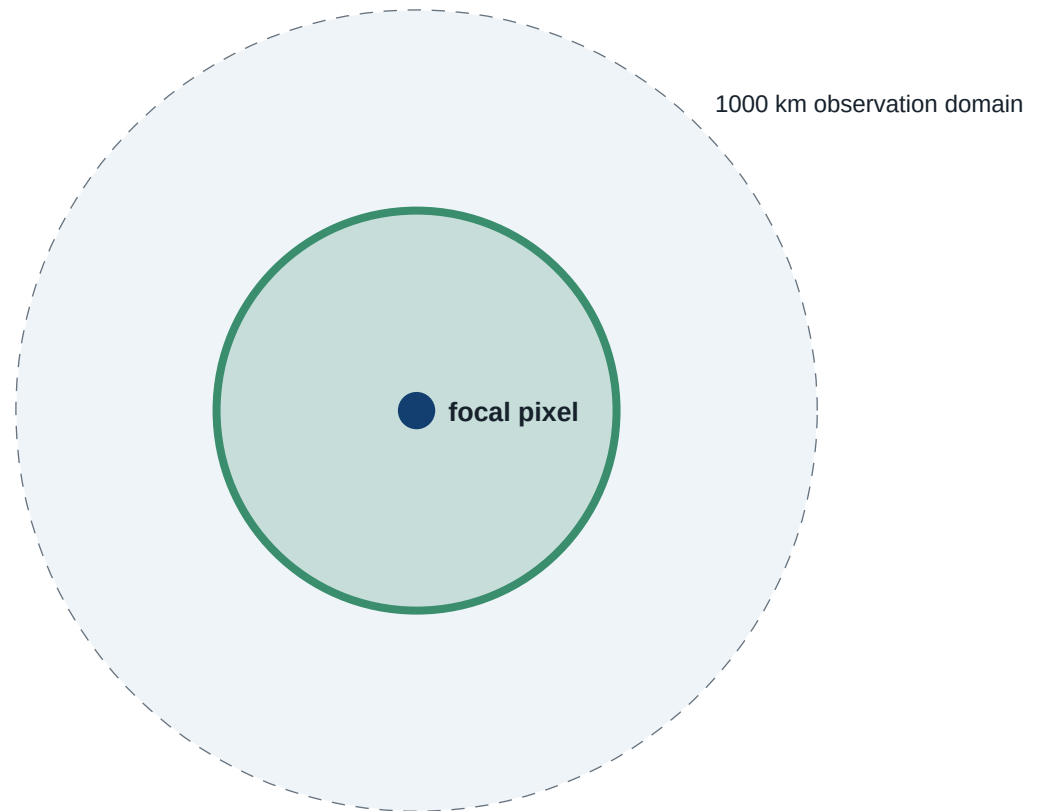
1. Every neighbor retains focal ID, neighbor ID, coordinates, and distance.
2. Cold, warm, and Delta are calculated once per canonical edge.
3. Self-pairs are excluded before reduction.



## 4. The 1000 km observation domain

$D_{\max}$  is support, not the answer.

1.  $D_{\max} = 1000$  km standardizes the maximum view.
2.  $R^*$  is estimated inside that view or remains unresolved.
3. The workflow never substitutes 1000 km for a missing  $R^*$ .

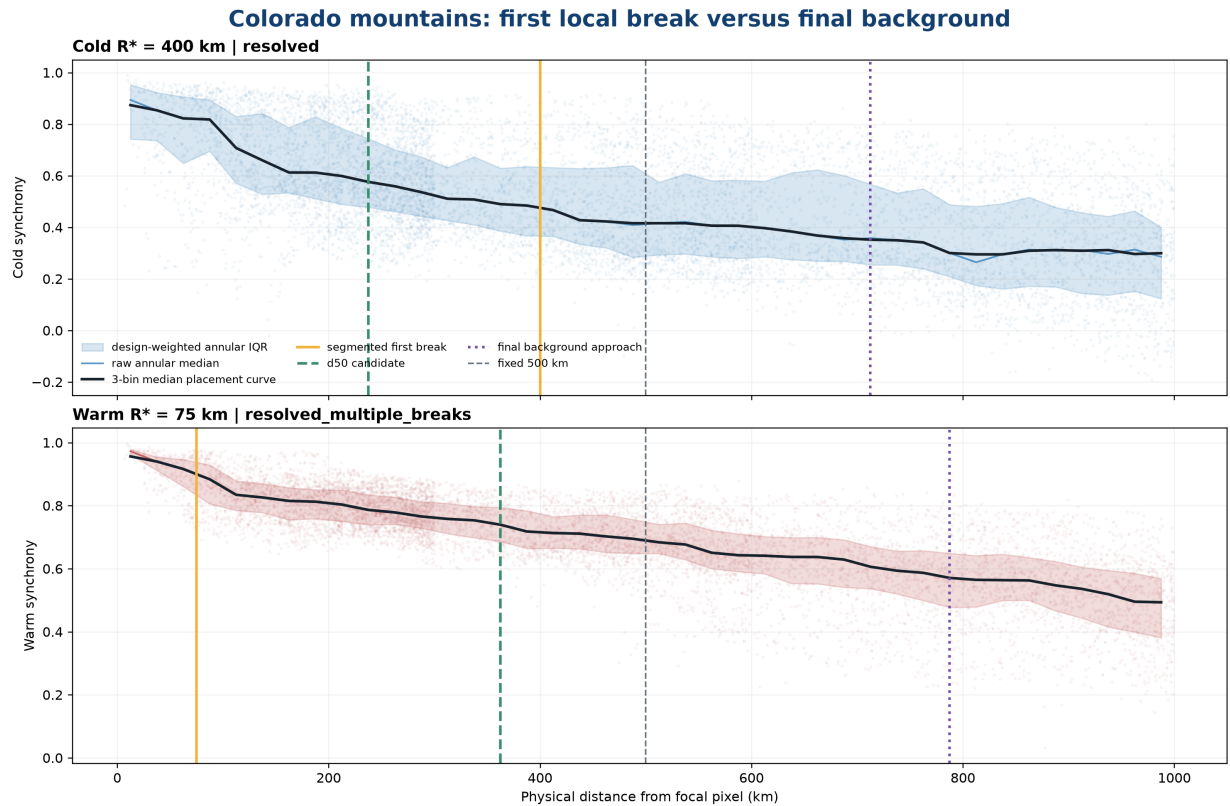


**$R^*$  unresolved - do not substitute 1000 km**

## 5. Pairwise synchrony versus distance

Real PRISM pairs reveal non-monotonic structure.

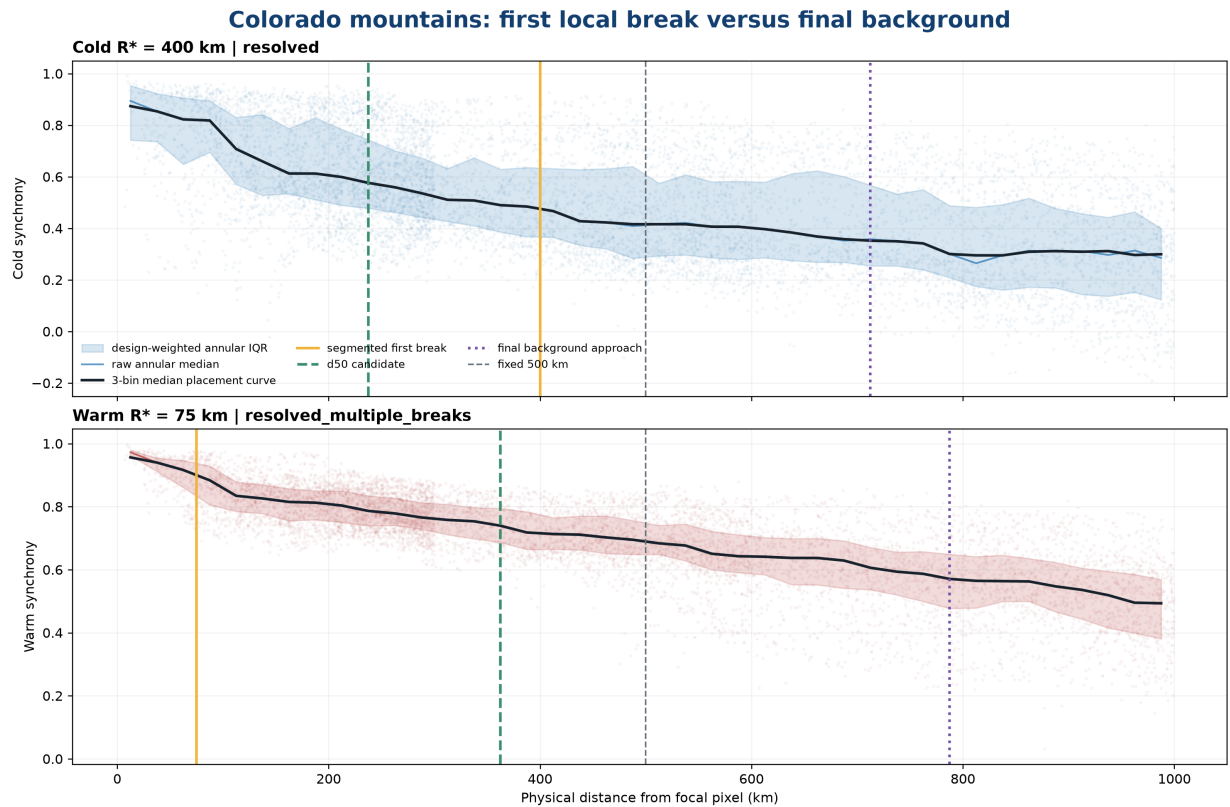
1. Five exhaustive focal runs retained 645,367 nonself relationships.
2. Raw pairs, annular medians, IQR, and smoothed placement curves remain visible.
3. Distance is great-circle kilometers.



## 6. Cold empirical decay curve

Lower-tail TMIN is evaluated independently.

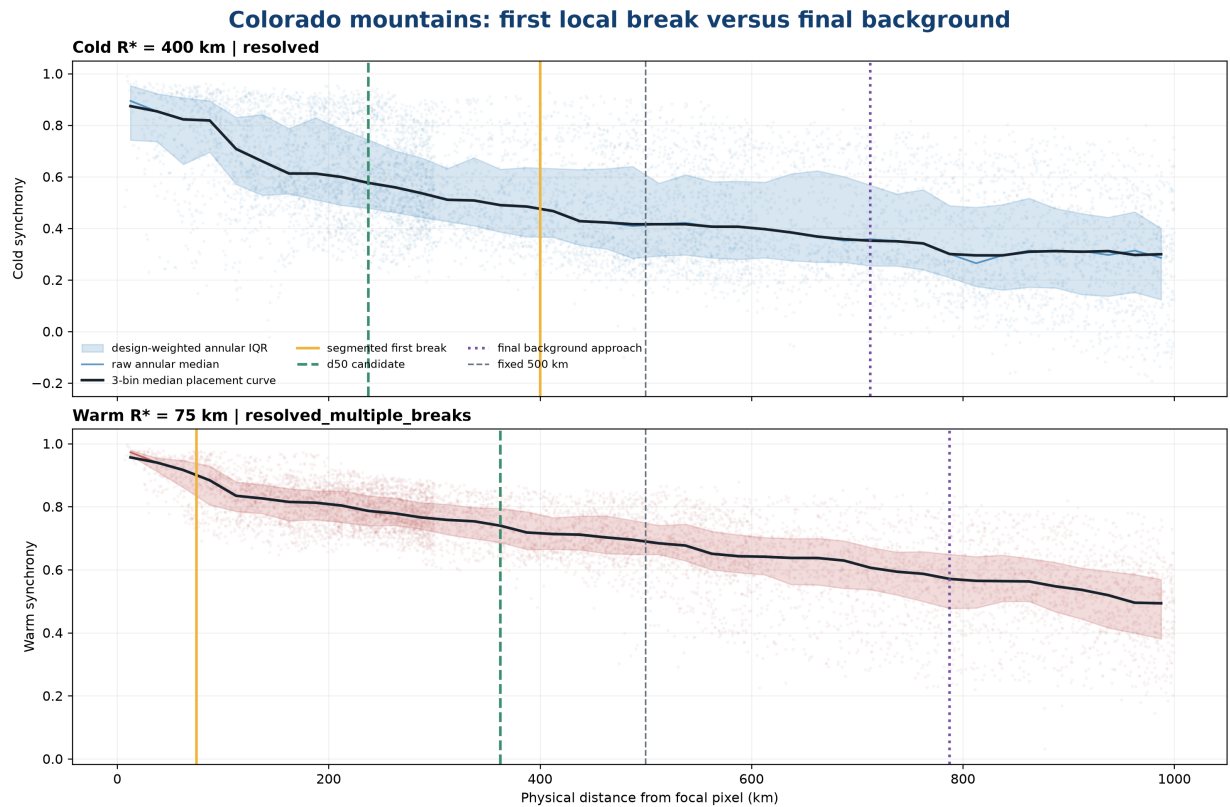
1. The raw annular curve is not forced to be monotonic.
2. The first break is separated from the final background approach.
3. Near and far robust slopes must differ enough to support a break.



## 7. Warm empirical decay curve

Strict upper-tail TMAX keeps the validated semantics.

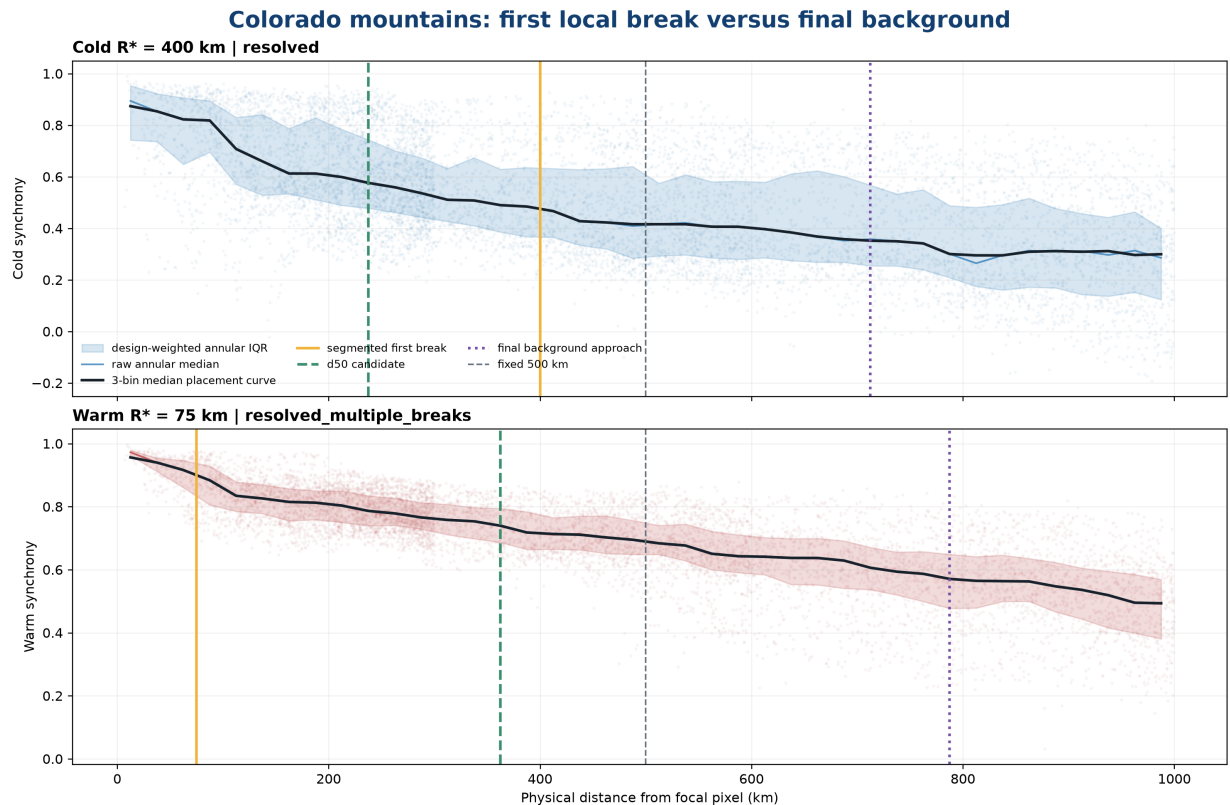
1. Warm uses observations strictly above pair-valid medians.
2. Warm break validity is not borrowed from cold.
3. An ambiguous or censored warm result blocks the common radius.



## 8. Local, regional, and broad-scale structure

Multiple scales are retained rather than compressed silently.

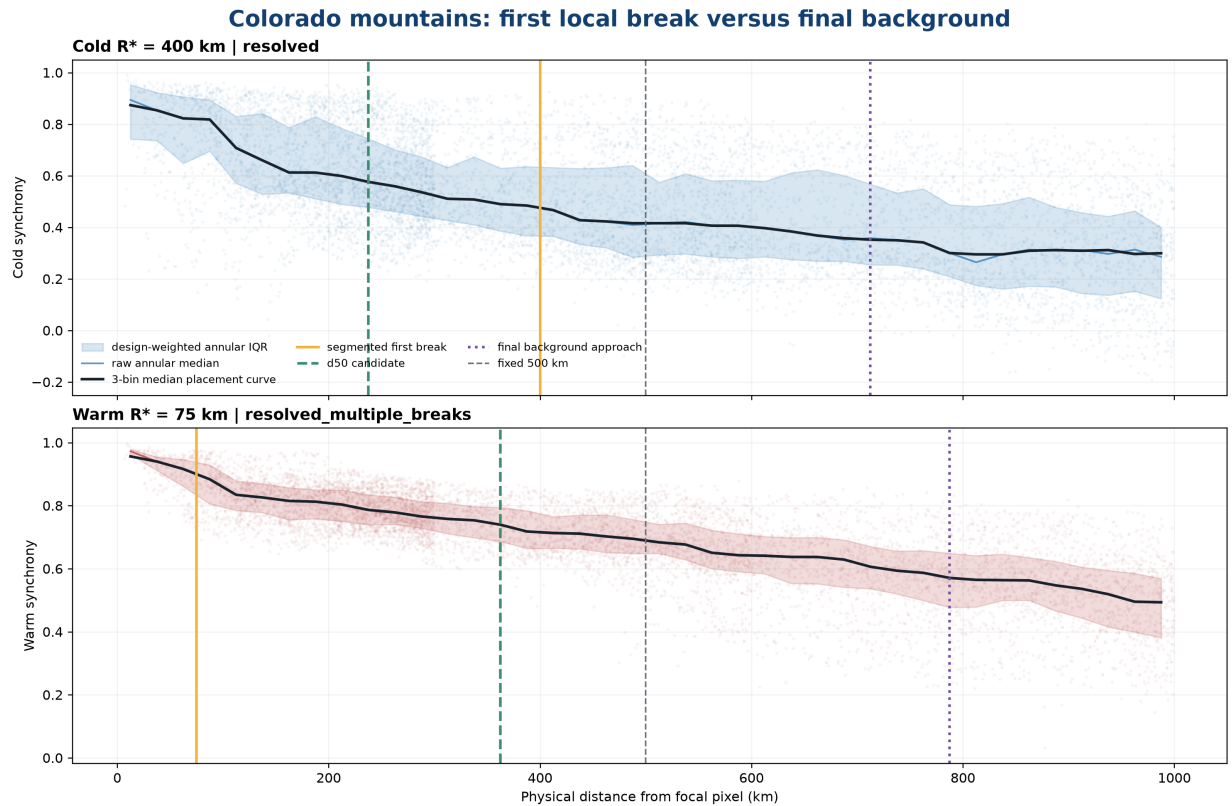
1. The segmented candidate targets the first steep-to-flatter transition.
2. A later transition is recorded as multiple-break structure.
3. Final background approach remains a separate diagnostic.



## 9. Detect cold $R^*$

A robust two-segment candidate must beat one robust line.

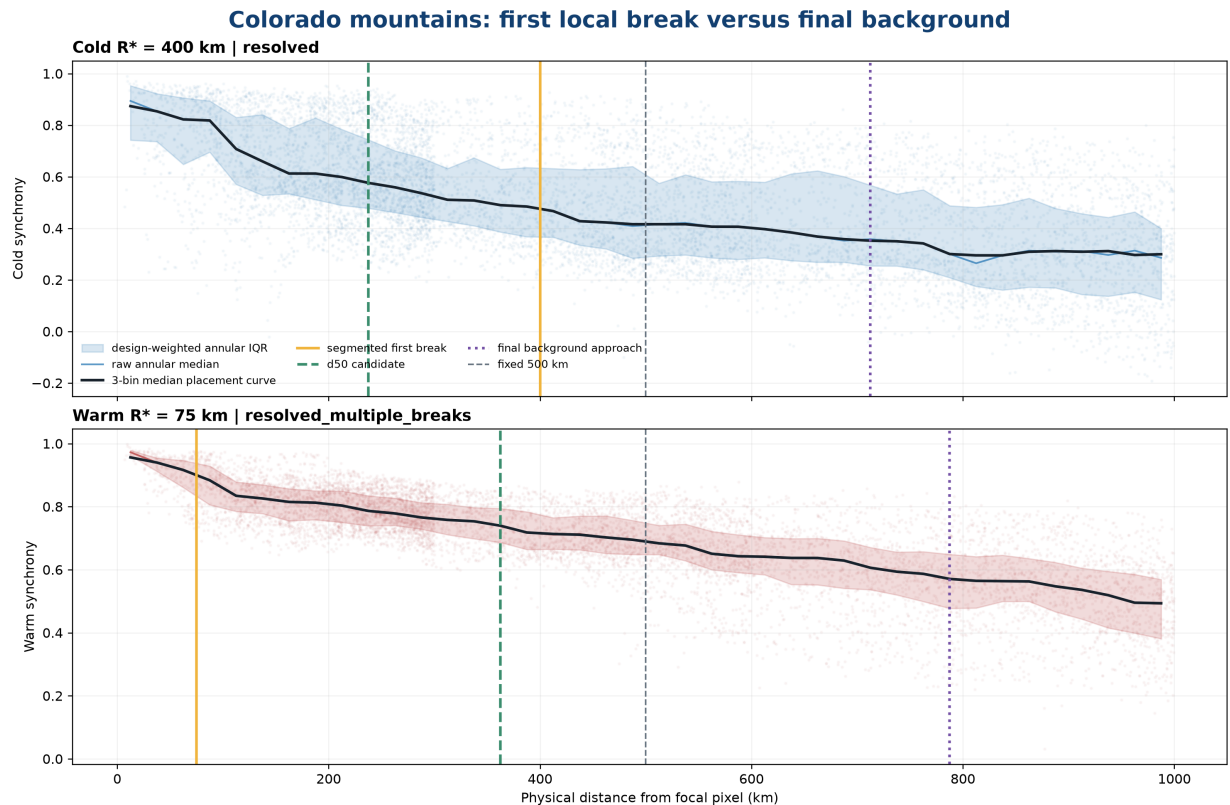
1. Candidate placement uses a three-bin centered median only.
2. Raw annular medians and IQR remain the evidence.
3. Near-field decay, slope change, improvement, ambiguity, and boundary support are checked.



## 10. Detect warm $R^*$

The same estimator and thresholds are applied to warm synchrony.

1. No method is selected because it produces more valid pixels.
2. d50, knee, derivative, and background-approach distances remain comparison candidates.
3. Sampling-induced status changes count as failures.

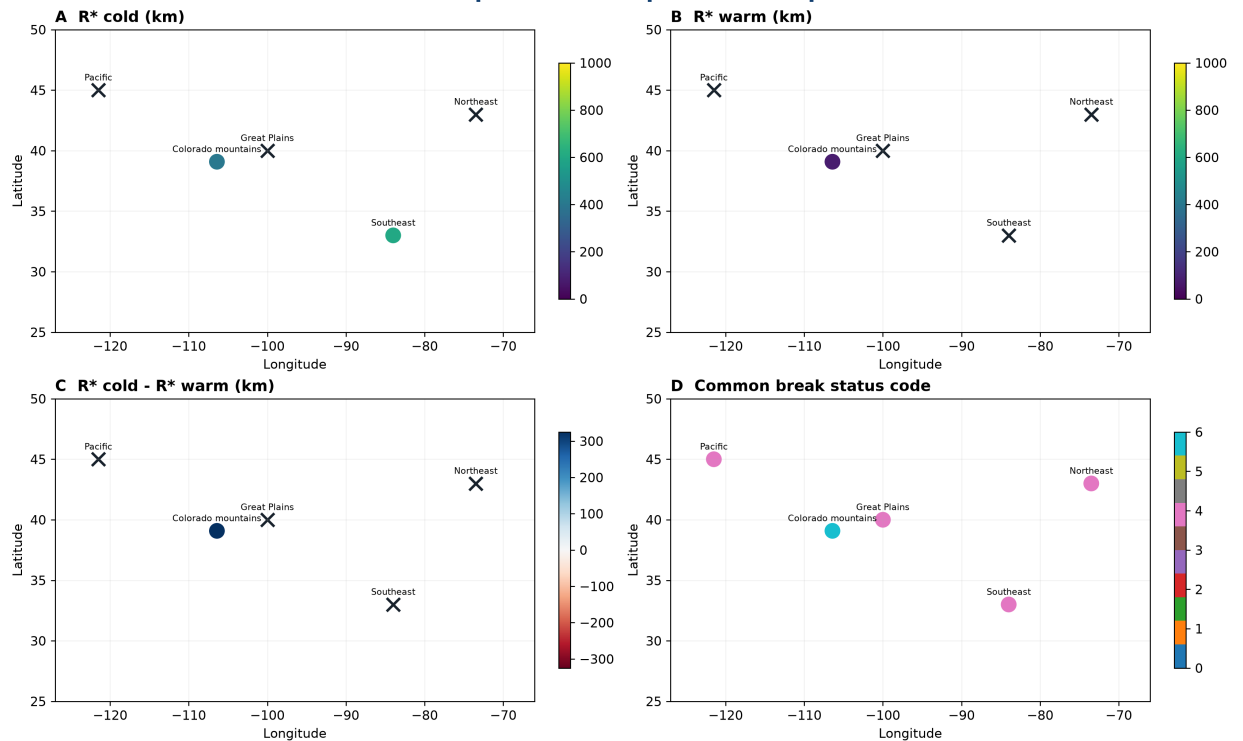


# 11. Define common $R^*$

Cold and warm use one shared adaptive neighbor set.

1. Default rule:  $\max(R^*_{\text{cold}}, R^*_{\text{warm}})$ .
2. The rule is configurable but not changed post hoc.
3. Only 1/5 sampled representatives had a valid common  $R^*$ .

Scale outputs at five representative pixels

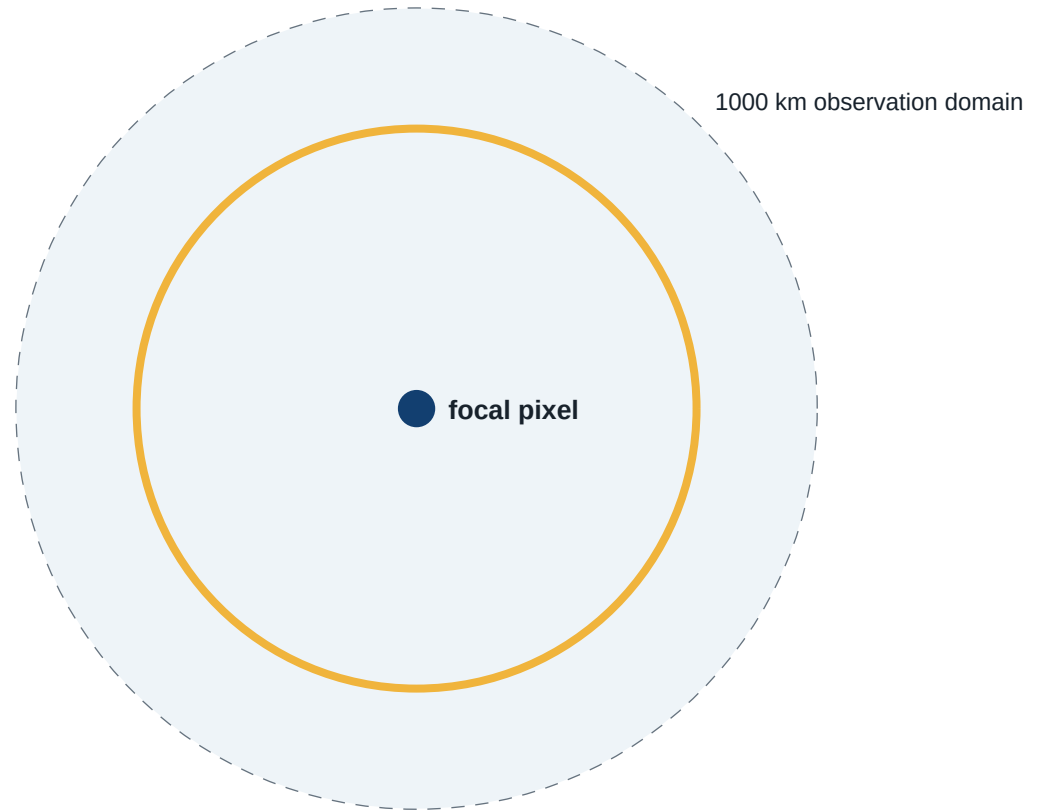


REPRESENTATIVE GATE FAILED - COLORADO STAGE WITHHELD

## 12. Fixed 500 km neighborhood

The comparison control comes from the same 1000 km pair product.

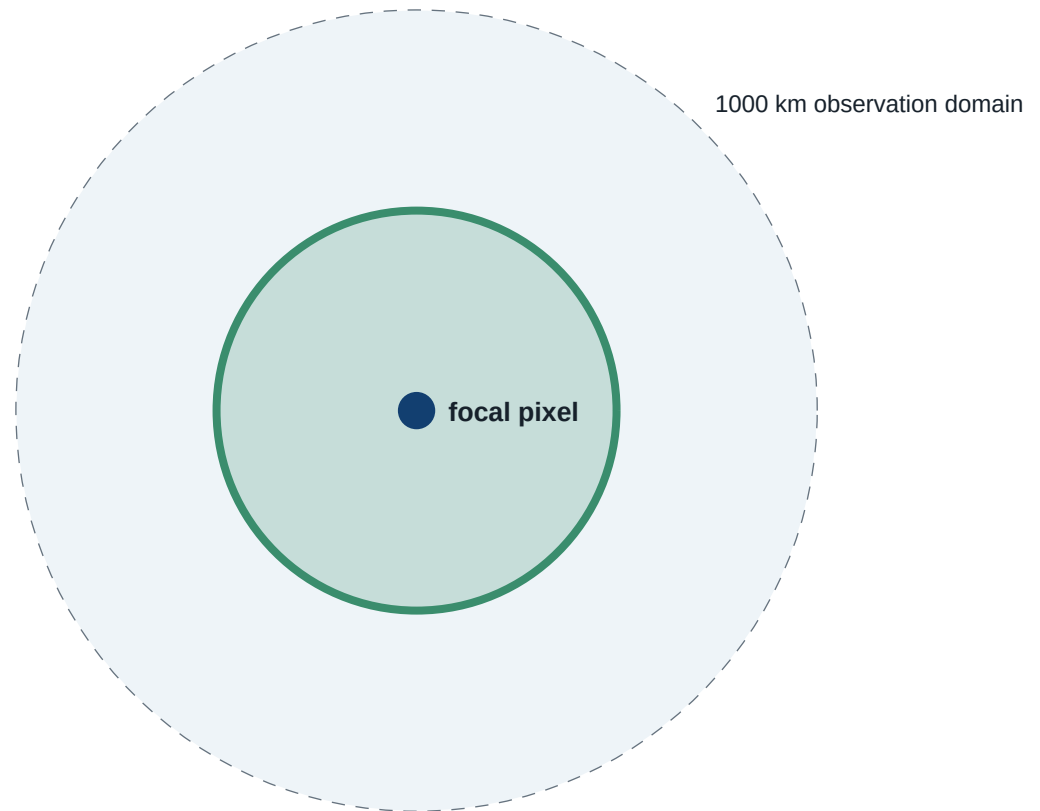
1. No pair correlation is recalculated for the 500 km baseline.
2. Design-weighted medians estimate the complete relationship population.
3. Sampled fixed-Delta MAE versus exhaustive was 0.000651.



## 13. Adaptive $R^*$ neighborhood

Adaptive reduction is allowed only when both tails resolve.

1. Neighbors satisfy distance  $\leq R^*_{\text{common}}$ .
2. Cold and warm use the identical neighbor identities.
3. Unresolved, unstable, or ambiguous results remain missing.



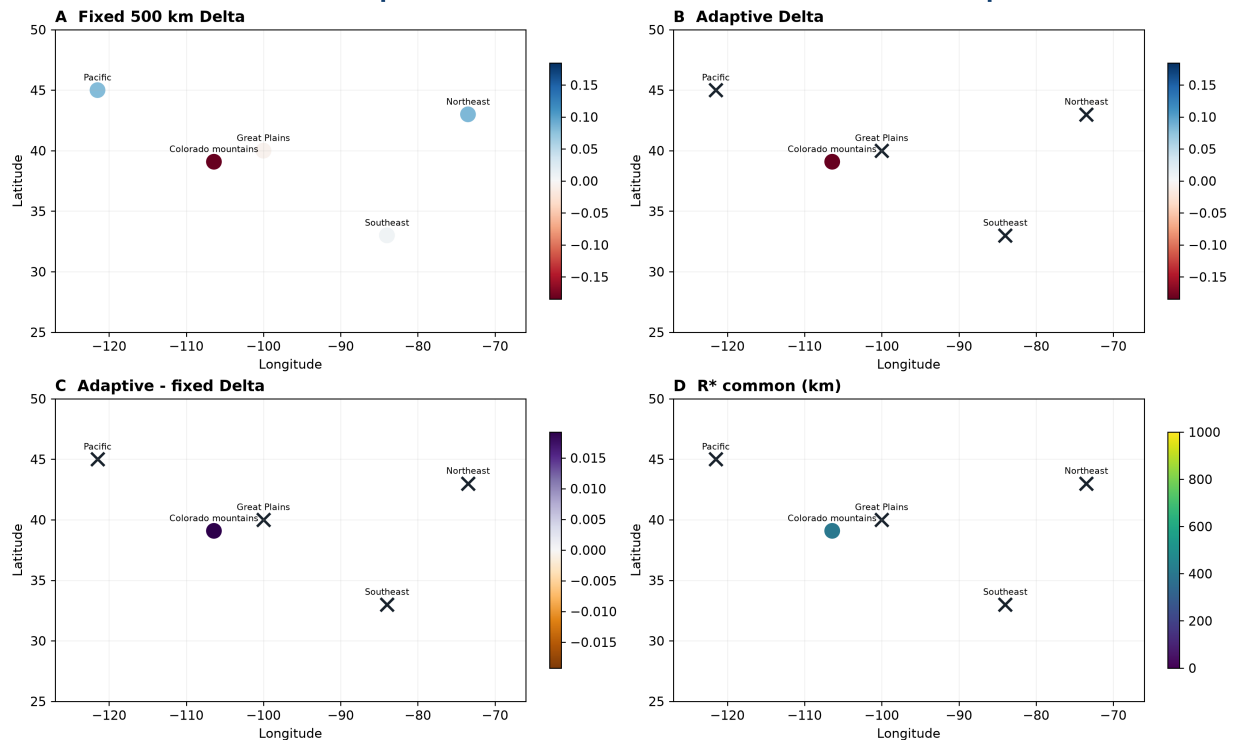
ONLY VALID REPRESENTATIVE PIXELS ADVANCE

# 14. Adaptive cold collapse

Reduce cold pair values inside the common radius.

1. The estimator uses the same robust empirical median as the fixed control.
2. Inverse inclusion probabilities correct the stratified far-field sample.
3. Observed and estimated pair counts are retained.

Bounded representative result - not a Colorado or CONUS map



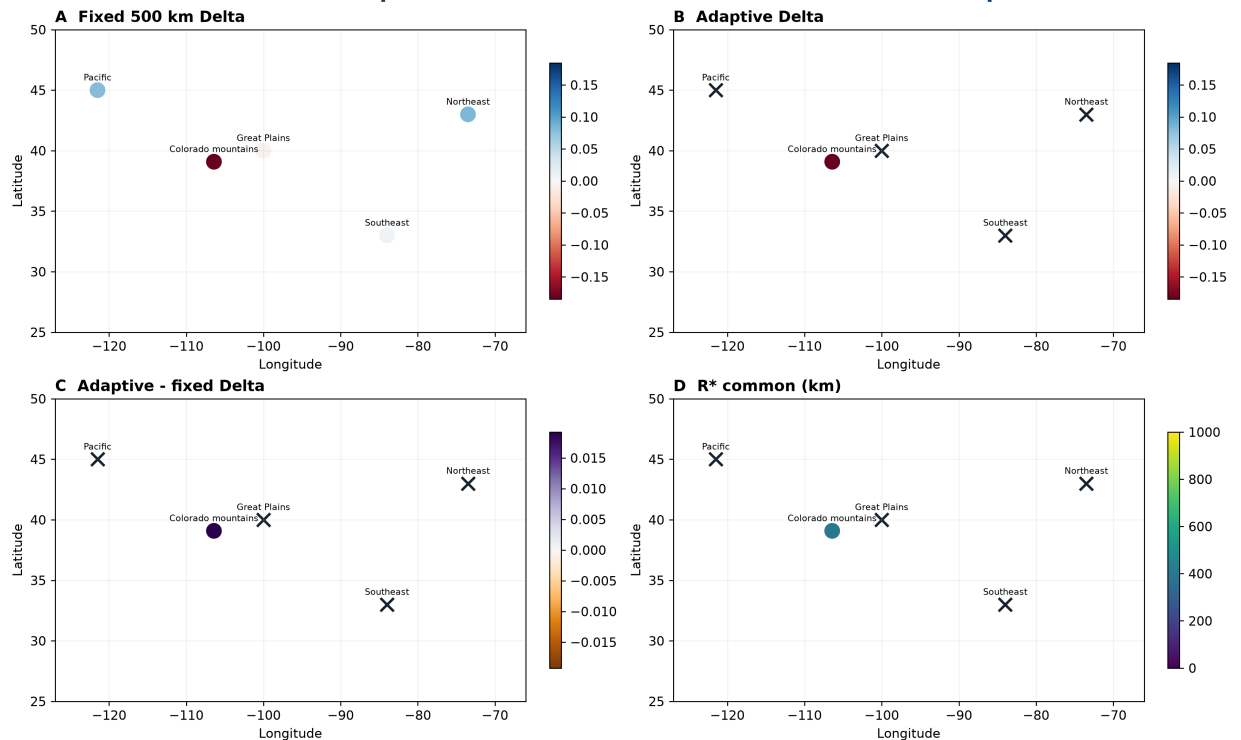
COLORADO MAP WITHHELD BY REPRESENTATIVE GATE

# 15. Adaptive warm collapse

Warm strength changes only through neighbor inclusion.

1. The tail definition, correlation statistic, and reduction do not change.
2. The same common radius avoids strength-versus-extent confounding.
3. Invalid common breaks produce no adaptive warm estimate.

**Bounded representative result - not a Colorado or CONUS map**



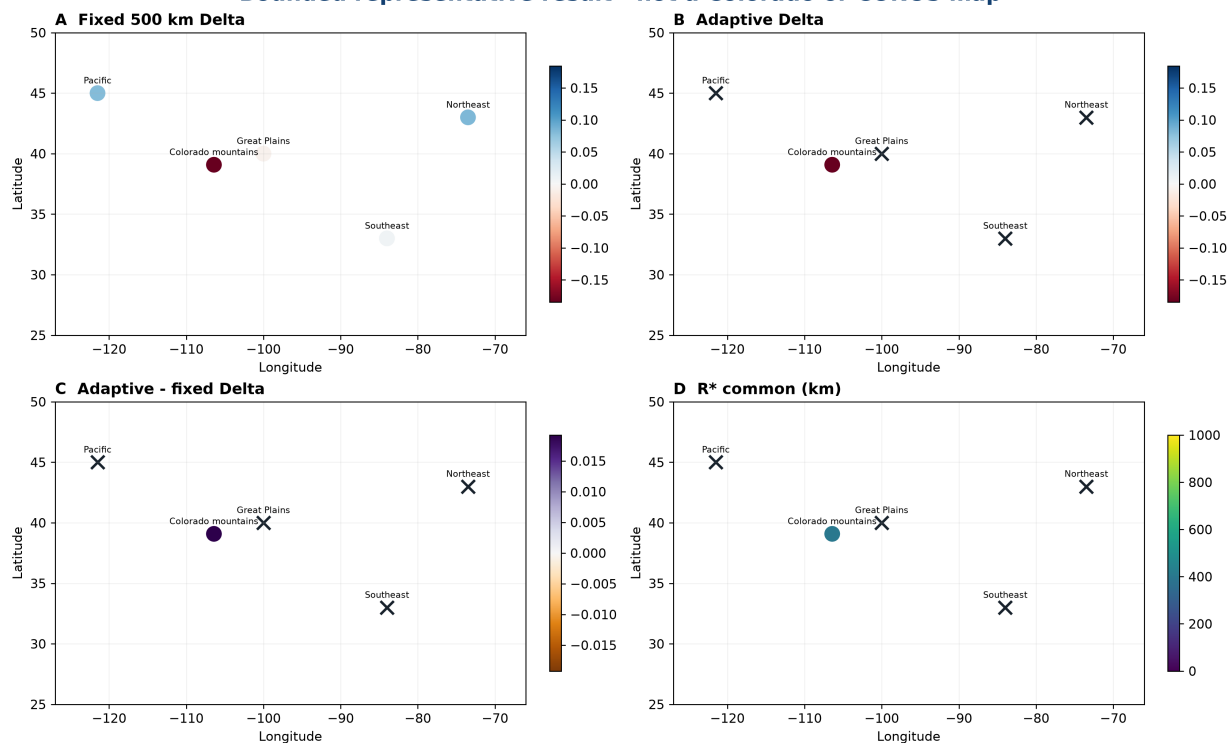
COLORADO MAP WITHHELD BY REPRESENTATIVE GATE

## 16. Adaptive Delta collapse

Pair identity is preserved through the final contrast.

1. For each retained neighbor:  $\Delta_{ij} = S_{\text{cold},ij} - S_{\text{warm},ij}$ .
2. Primary Delta is the weighted median of those paired differences.
3. The reduction gap remains a separate QC field.

Bounded representative result - not a Colorado or CONUS map

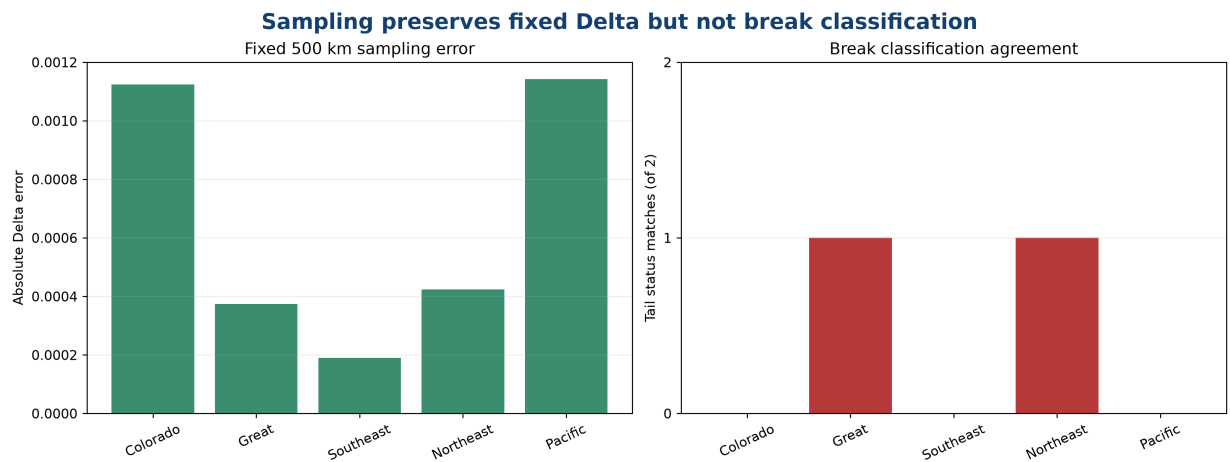


COLORADO MAP WITHHELD BY REPRESENTATIVE GATE

## 17. Repeat across pixels

The staged gate prevents an unsupported scale field.

1. Exhaustive common breaks resolved at 2/5 representatives.
2. Sampled common breaks resolved at 1/5 representatives.
3. The required 4/5 representative threshold was not met.



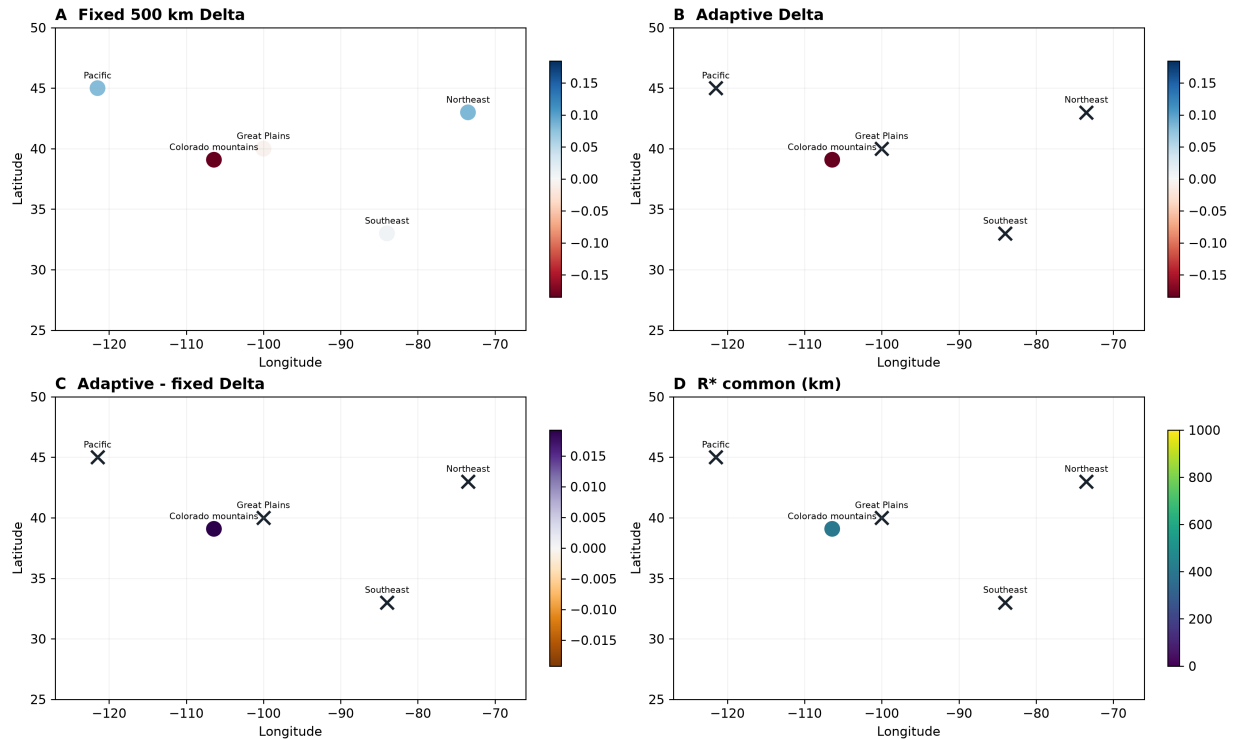
**STOP: NO 25-SITE COLORADO OR FULL-STATE RUN**

## 18. Fixed 500 km cold

Available at all five representative pixels.

1. This is a strength summary, not a scale estimate.
2. It is calculated from relationships within 500 km.
3. The figure shows representative points only.

Bounded representative result - not a Colorado or CONUS map

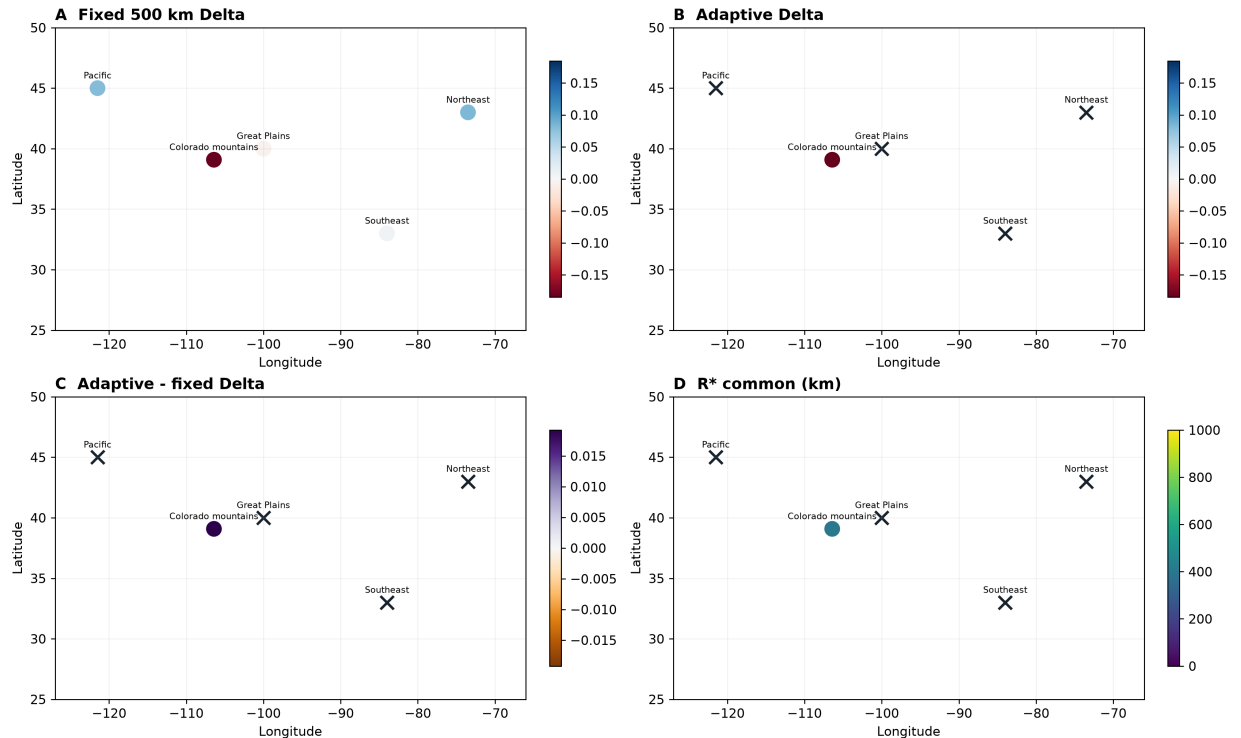


# 19. Adaptive cold

Available only where the common break is valid.

1. Missing symbols are scientific results, not plotting omissions.
2. The sampled estimator advanced only one common site.
3. No spatial interpolation is used to hide missing support.

**Bounded representative result - not a Colorado or CONUS map**



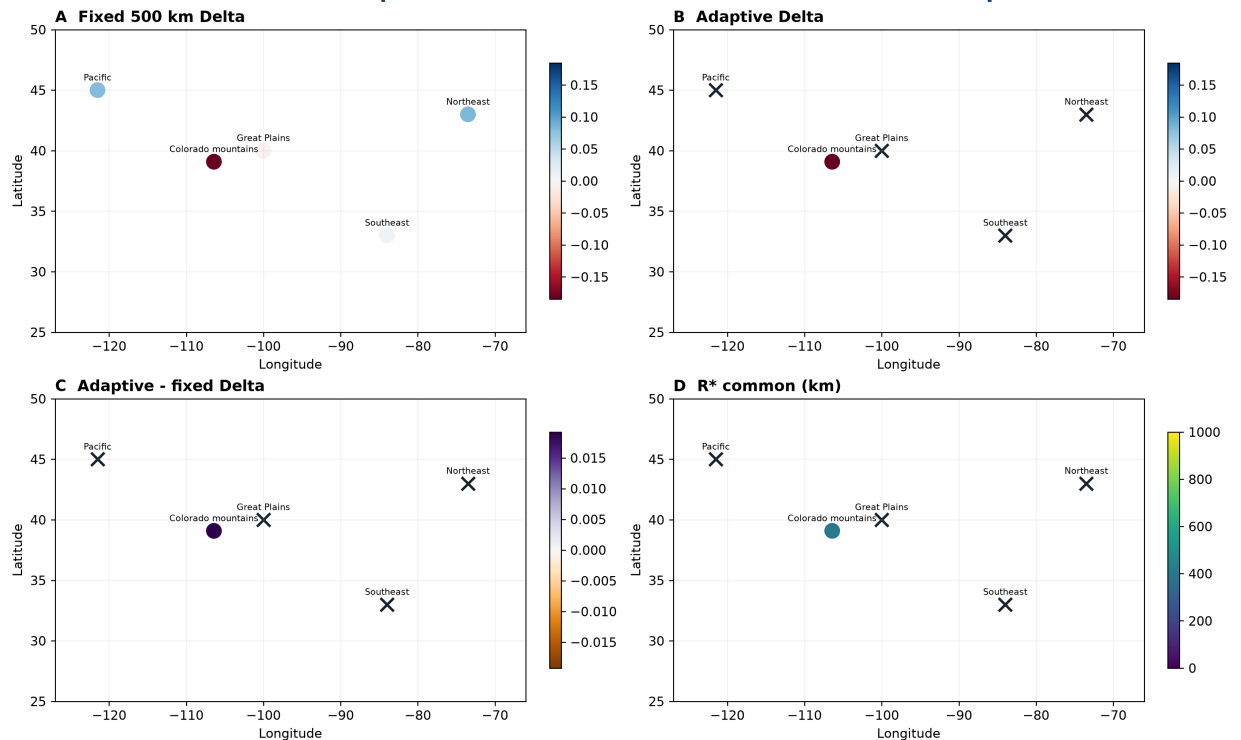
PRODUCTION MAP WITHHELD

## 20. Cold difference

Adaptive minus fixed requires both estimates.

1. The intended scale is zero-centered and diverging.
2. Representative sparsity prevents a defensible geographic surface.
3. The output schema and calculation are implemented and tested.

**Bounded representative result - not a Colorado or CONUS map**



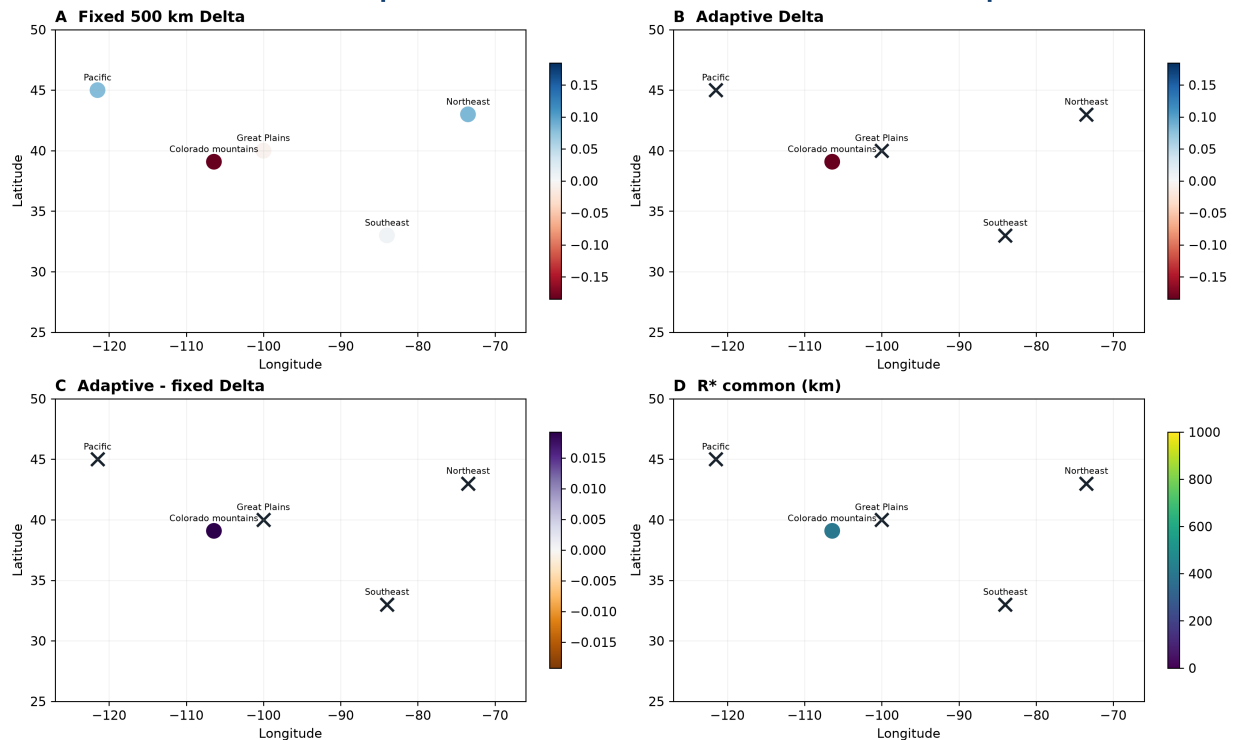
PRODUCTION MAP WITHHELD

## 21. Fixed 500 km warm

Strict upper-tail TMAX remains unchanged.

1. The fixed warm control is complete at the representatives.
2. Sampling error is small for the fixed collapse.
3. This does not validate sampled break classification.

Bounded representative result - not a Colorado or CONUS map

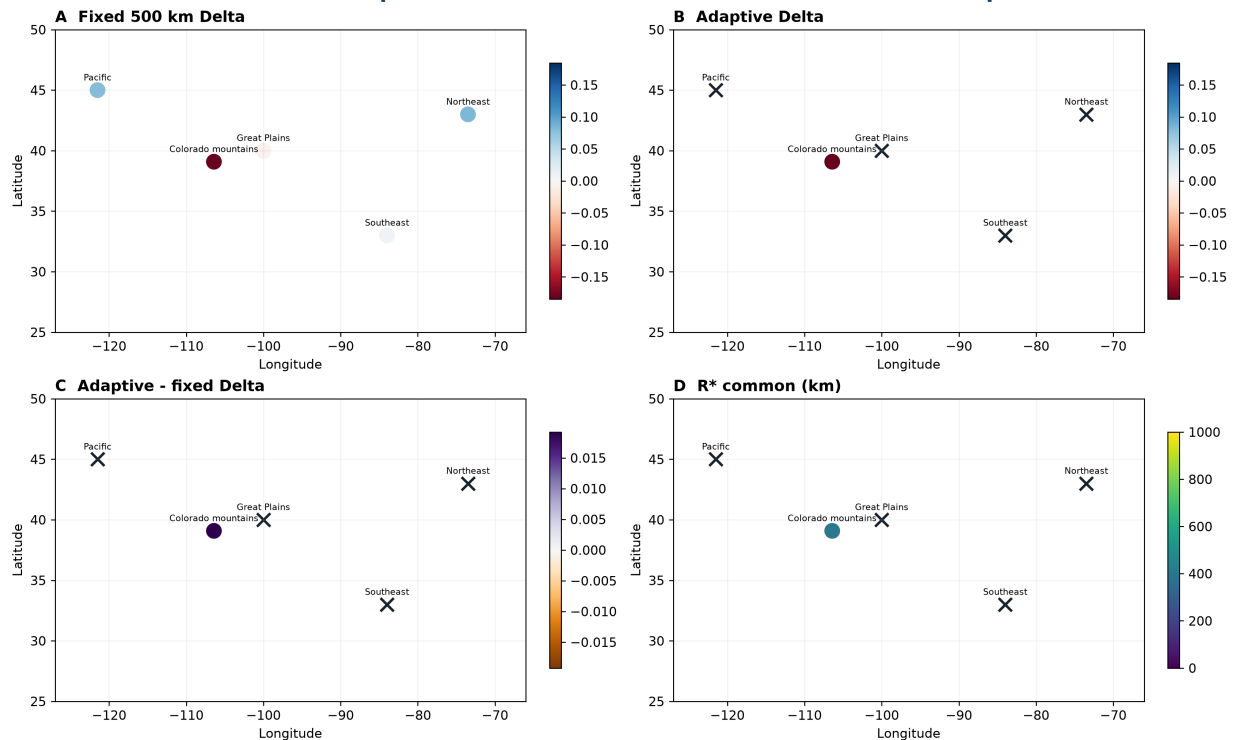


## 22. Adaptive warm

A valid common radius is required.

1. Cold and warm are not allowed to use different neighbor sets.
2. Ambiguous warm breaks block the common result.
3. This conservative rule prevents artificial Delta changes.

Bounded representative result - not a Colorado or CONUS map



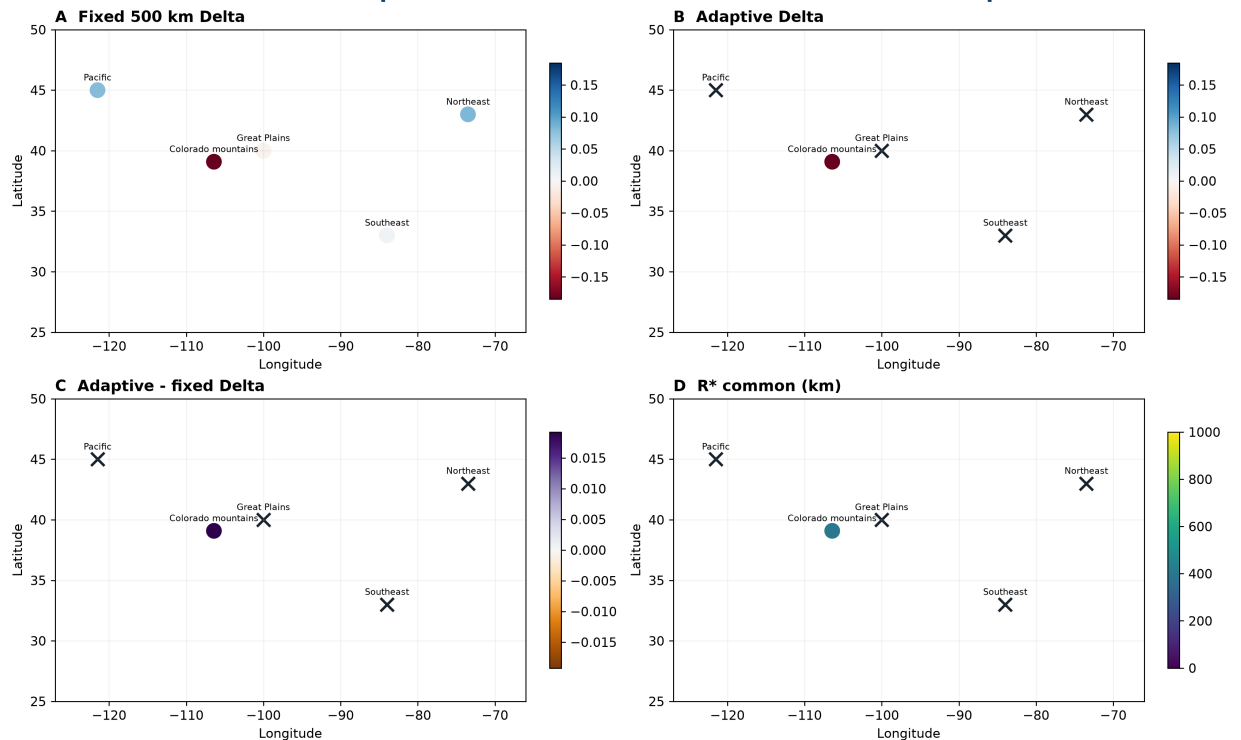
PRODUCTION MAP WITHHELD

## 23. Warm difference

Adaptive minus fixed is retained where calculable.

1. Thresholds of 0.01, 0.025, and 0.05 are implemented in comparison statistics.
2. Too few common results exist for a stable spatial summary.
3. No visual claim is made from unavailable pixels.

Bounded representative result - not a Colorado or CONUS map



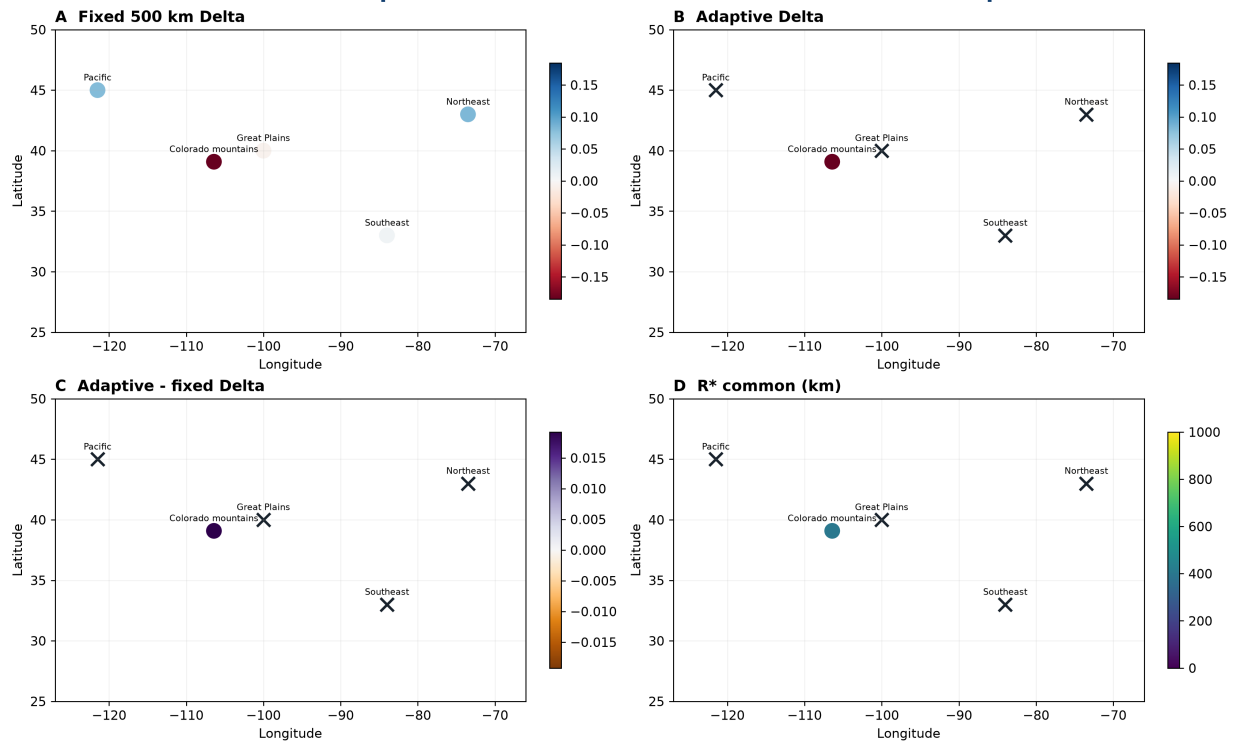
PRODUCTION MAP WITHHELD

## 24. Fixed 500 km Delta

Red means warm stronger; blue means cold stronger.

1. Delta > 0 means cold synchrony is stronger.
2. Delta < 0 means warm synchrony is stronger.
3. The red-white-blue convention is preserved.

Bounded representative result - not a Colorado or CONUS map

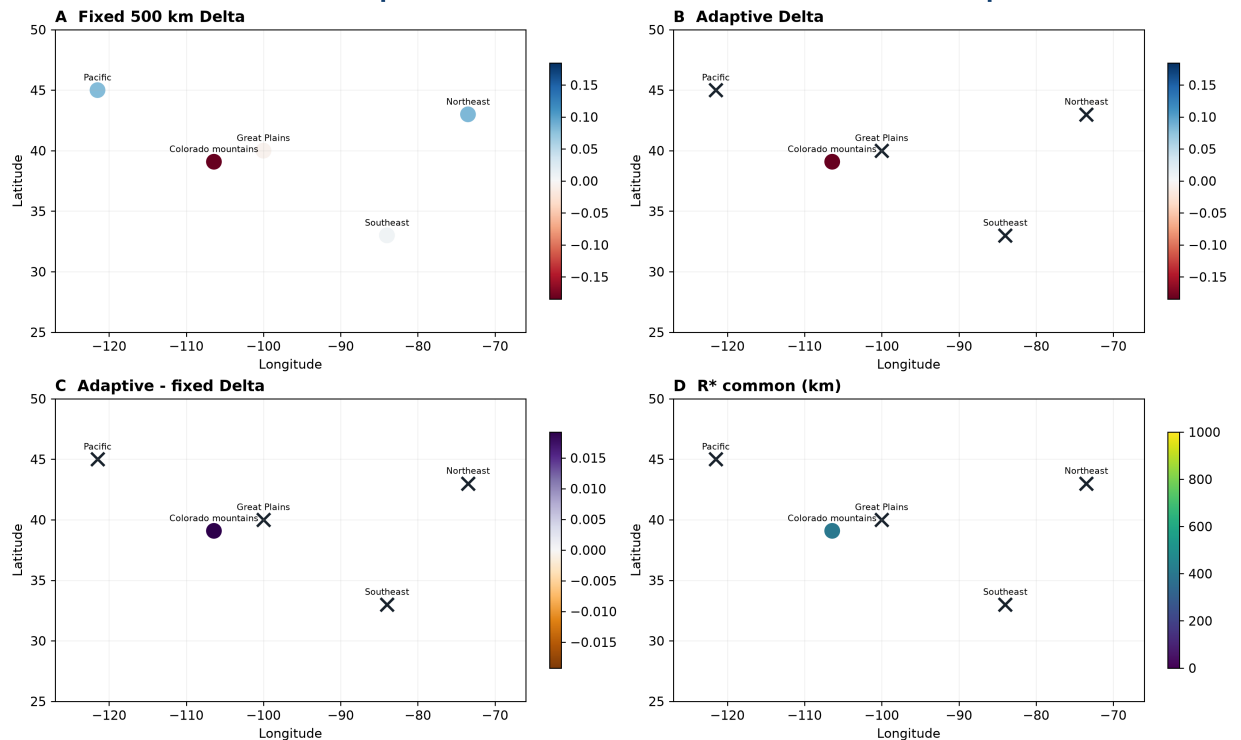


## 25. Adaptive Delta

The paired contrast uses the common adaptive domain.

1. No difference-of-marginal-medians substitution occurs.
2. The sign is scientifically interpretable only where  $R^*$  is valid.
3. Adaptive Delta is not extrapolated into unresolved pixels.

**Bounded representative result - not a Colorado or CONUS map**



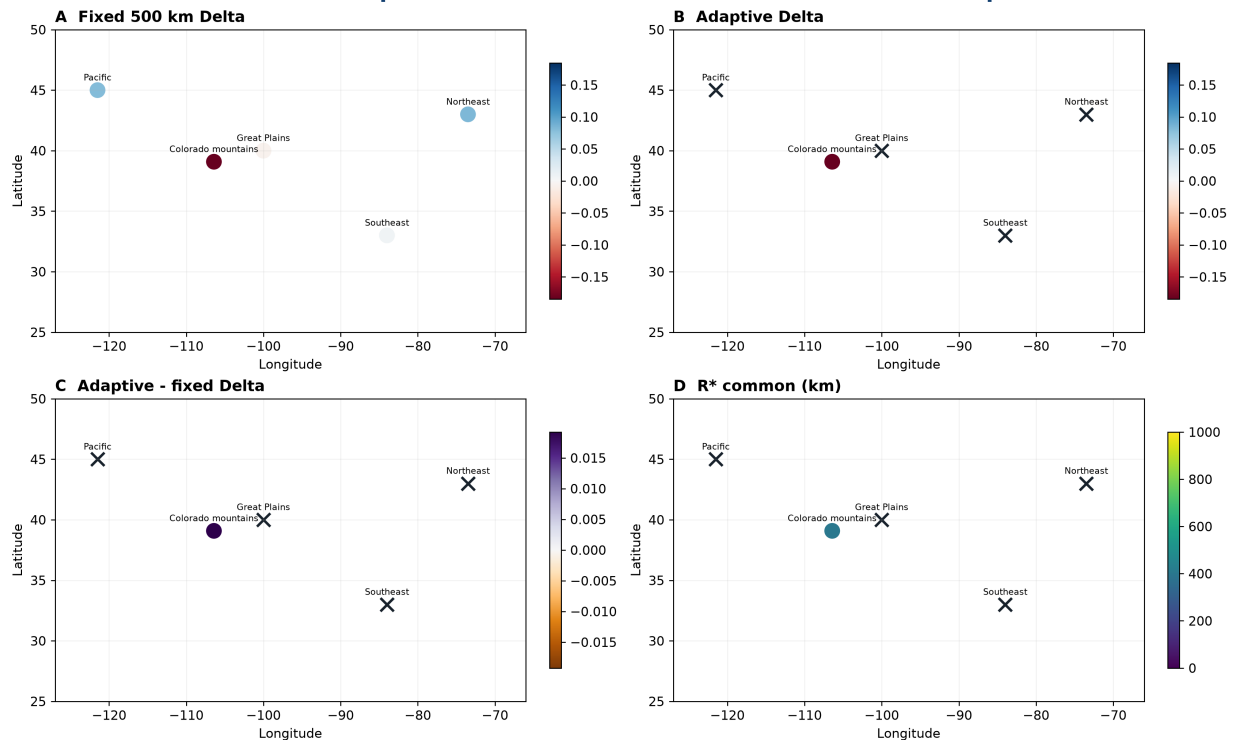
PRODUCTION MAP WITHHELD

## 26. Delta difference

A zero-centered comparison is ready but under-supported.

1. Difference = Delta\_adaptive - Delta\_500.
2. A positive difference is not automatically better.
3. The experiment asks where inference changes, not whether it improves.

Bounded representative result - not a Colorado or CONUS map



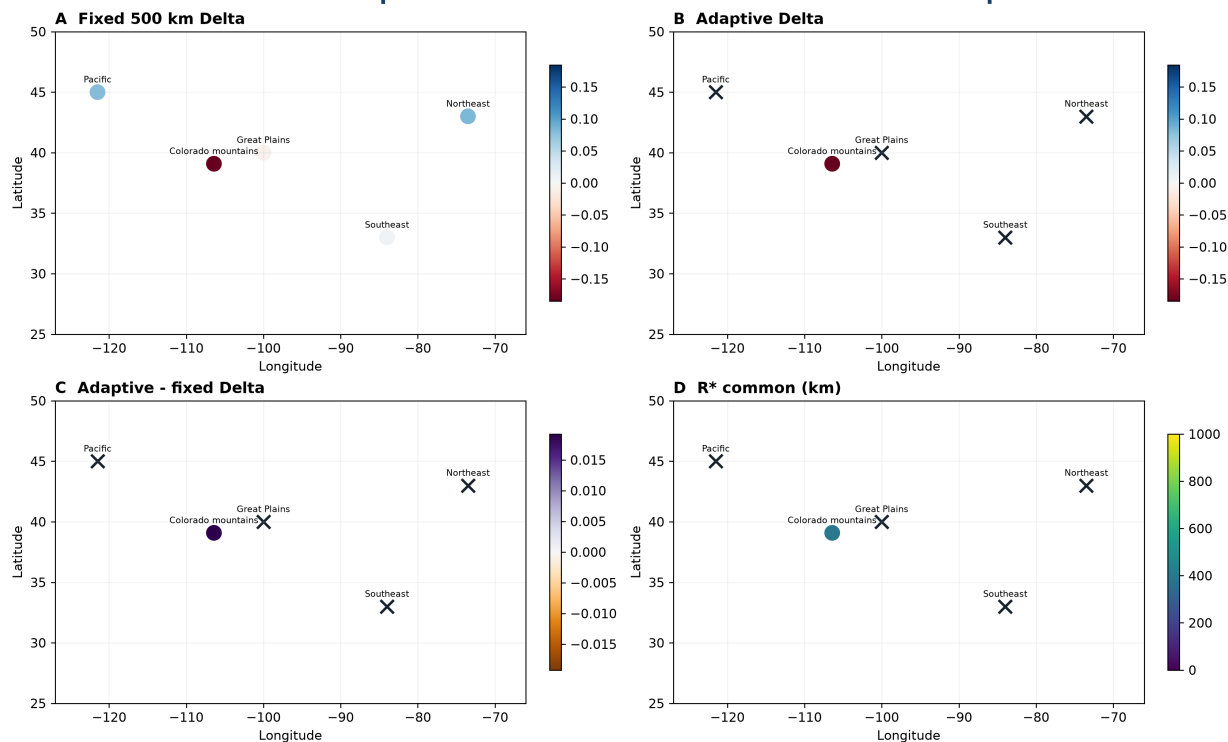
PRODUCTION MAP WITHHELD

## 27. Delta sign-change map

Qualitative inference changes require valid paired estimates.

1. A sign change switches cold-stronger versus warm-stronger interpretation.
2. There were zero sites with comparable exhaustive and sampled adaptive Delta.
3. A sign-change fraction is therefore not reported as a map result.

Bounded representative result - not a Colorado or CONUS map



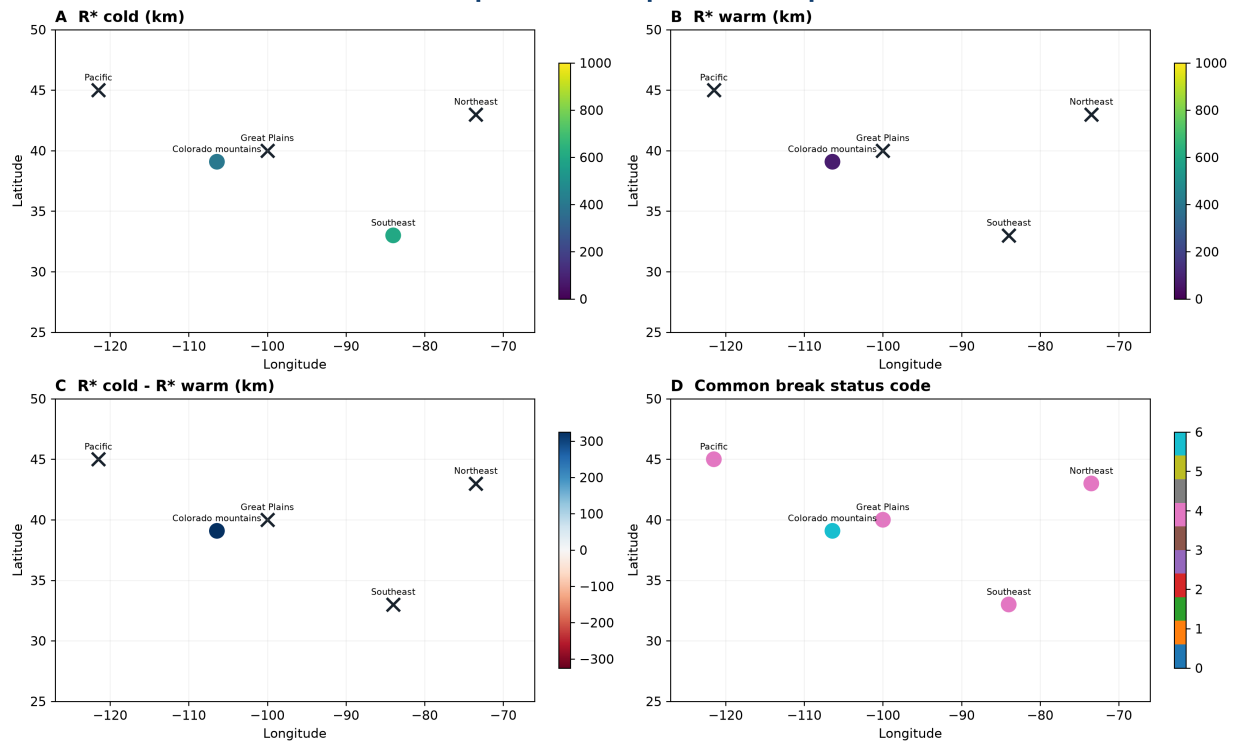
INSUFFICIENT VALID COMMON BREAKS

## 28. R\*\_cold

Cold characteristic distance is a separate scientific output.

1. Resolved, ambiguous, censored, and multiple-break states remain explicit.
2. The sequential palette represents kilometers, not synchrony strength.
3. The five-point panel is diagnostic, not a continuous map.

Scale outputs at five representative pixels

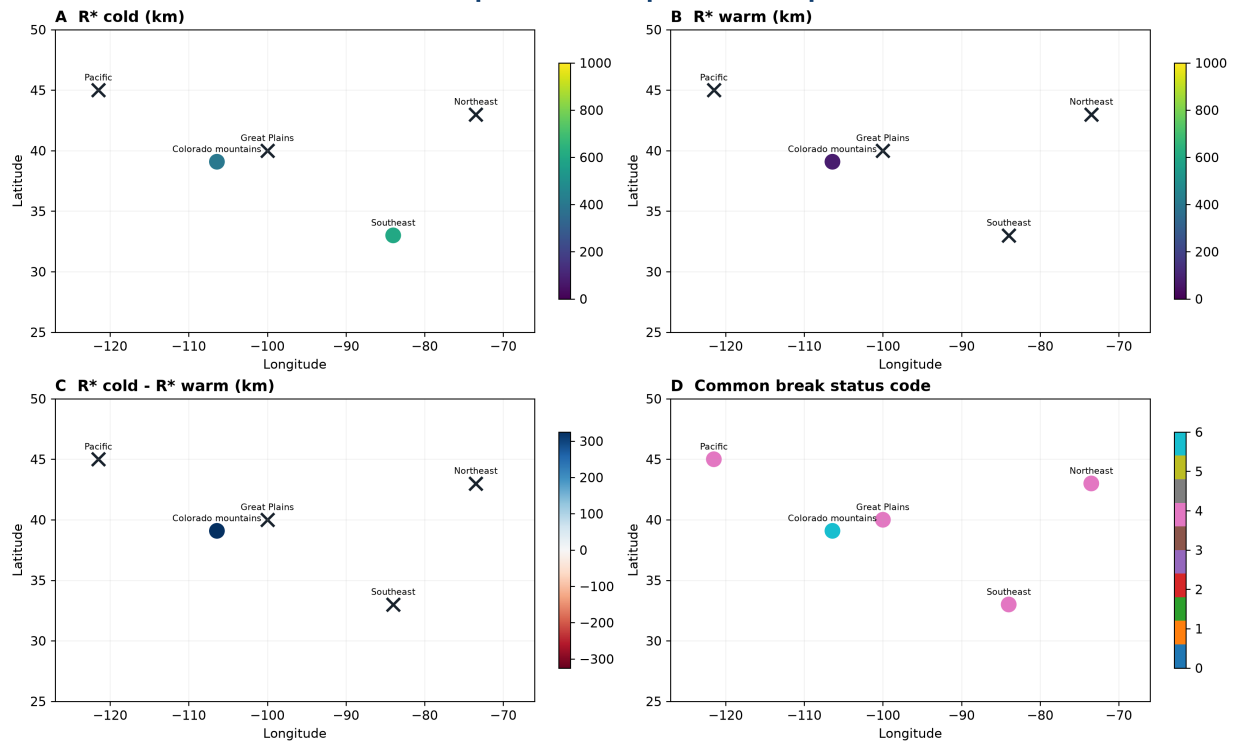


## 29. R\*\_warm

Warm characteristic distance is estimated independently.

1. Warm cannot inherit a cold break.
2. Many warm curves were ambiguous or boundary-limited.
3. That failure is central evidence for the HOLD decision.

Scale outputs at five representative pixels

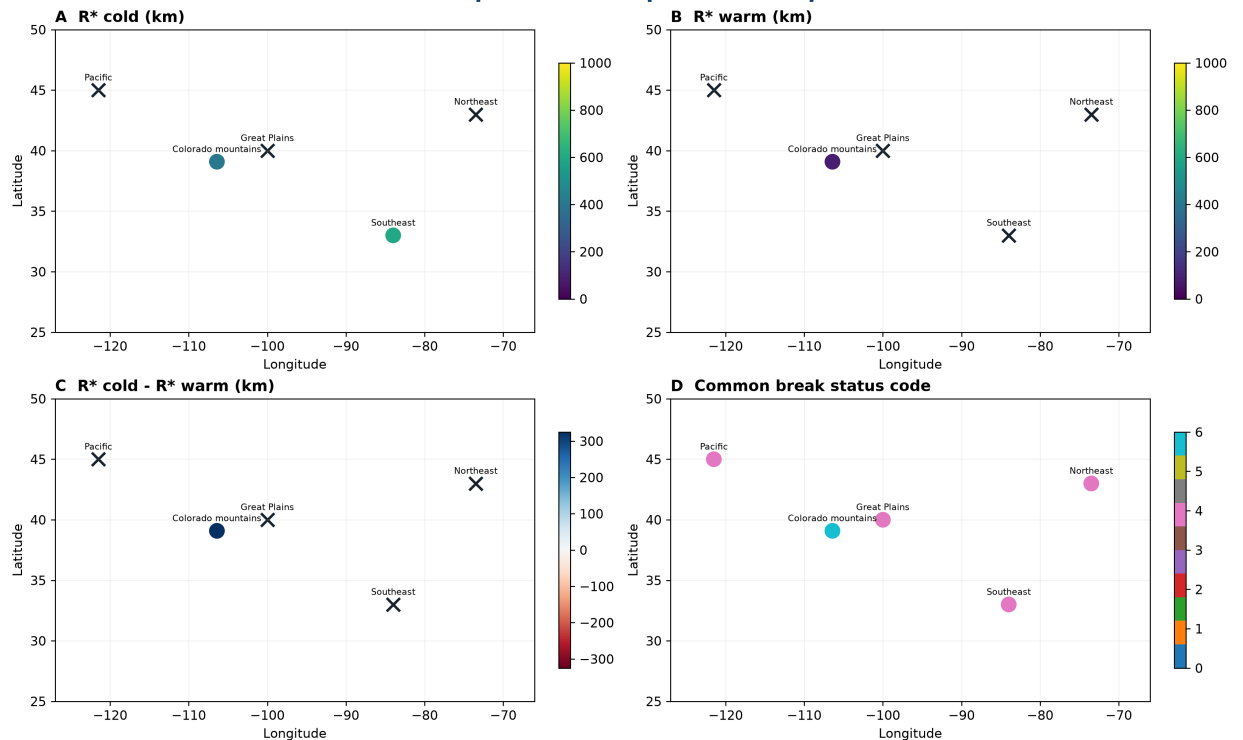


## 30. R\*\_common

The maximum is formed only when both components are valid.

1. R\*\_common is not set to 1000 km when unresolved.
2. Only valid common distances can drive adaptive reductions.
3. No statewide scale surface was produced.

Scale outputs at five representative pixels



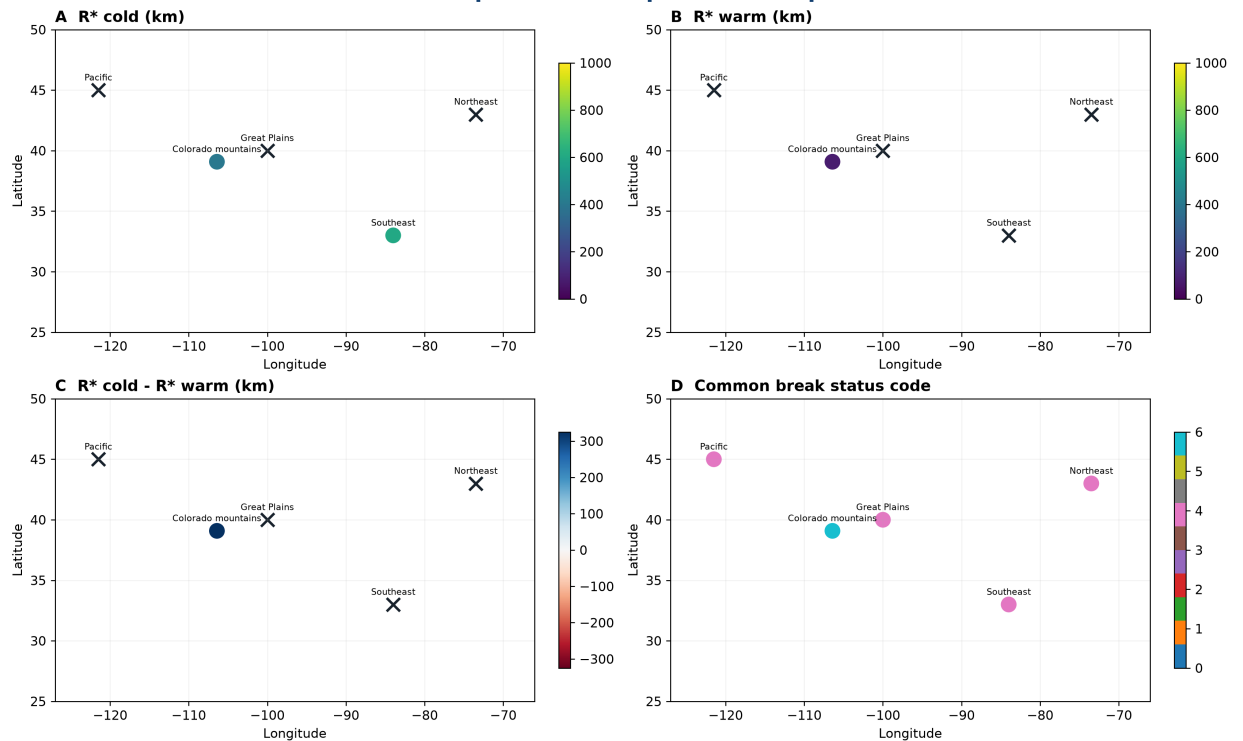
COLORADO SCALE MAP WITHHELD

## 31. Cold-minus-warm distance contrast

Delta\_R is not Delta\_S.

1.  $\Delta_R > 0$  means cold synchrony extends farther.
2.  $\Delta_R < 0$  means warm synchrony extends farther.
3. The contrast remains missing unless both distances are valid.

Scale outputs at five representative pixels

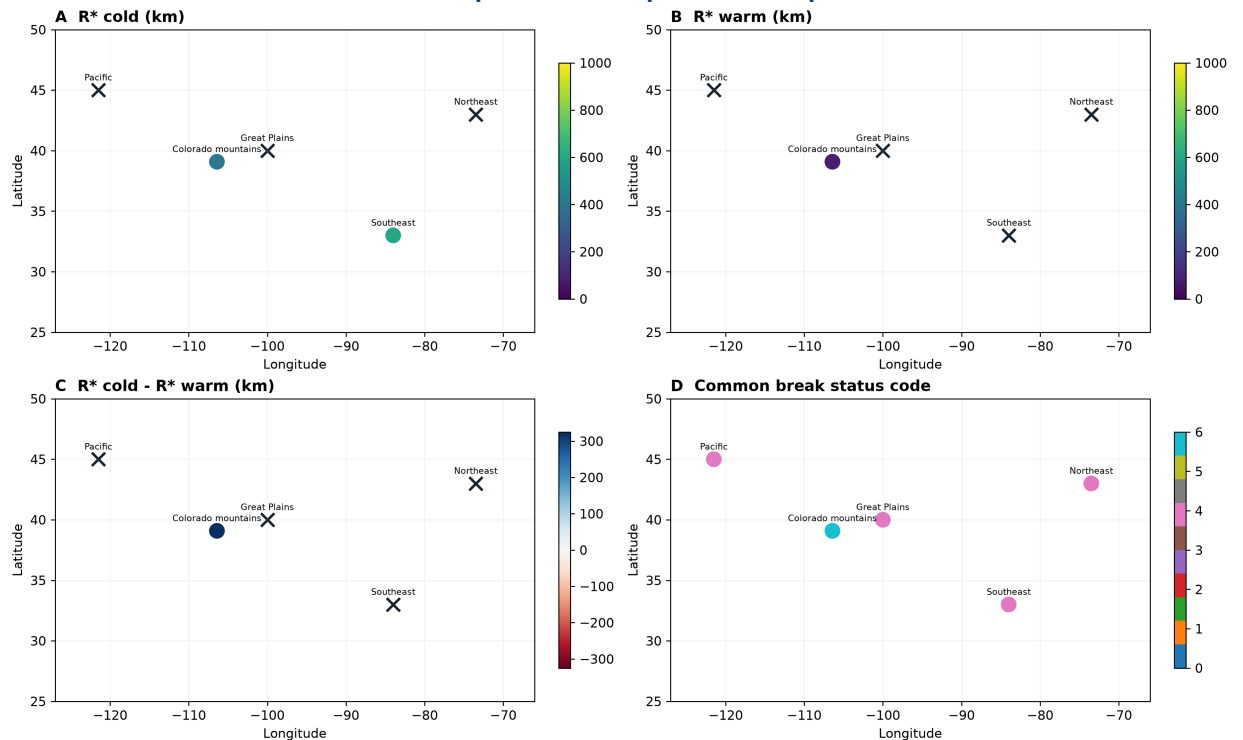


## 32. Unresolved and stability status

QC is mapped separately from continuous distance.

1. Status codes distinguish insufficient, flat, censored, unstable, ambiguous, resolved, and multiple breaks.
2. Crosses in the representative panels identify missing continuous values.
3. Unresolved areas are never hidden inside the sequential palette.

Scale outputs at five representative pixels



BREAK CLASSIFICATION AGREEMENT = 30%

## 33. Synchrony strength versus synchrony scale

These answer different scientific questions.

1. Strength summarizes covariance within a chosen domain.
2. Scale locates a reproducible transition in the distance response.
3. Neither is automatically a dispersal kernel, natural radius, or correlation length.

### SYNCHRONY STRENGTH

How strongly does the focal pixel covary with neighbors?

$$S(x, y, t)$$

### SYNCHRONY SCALE

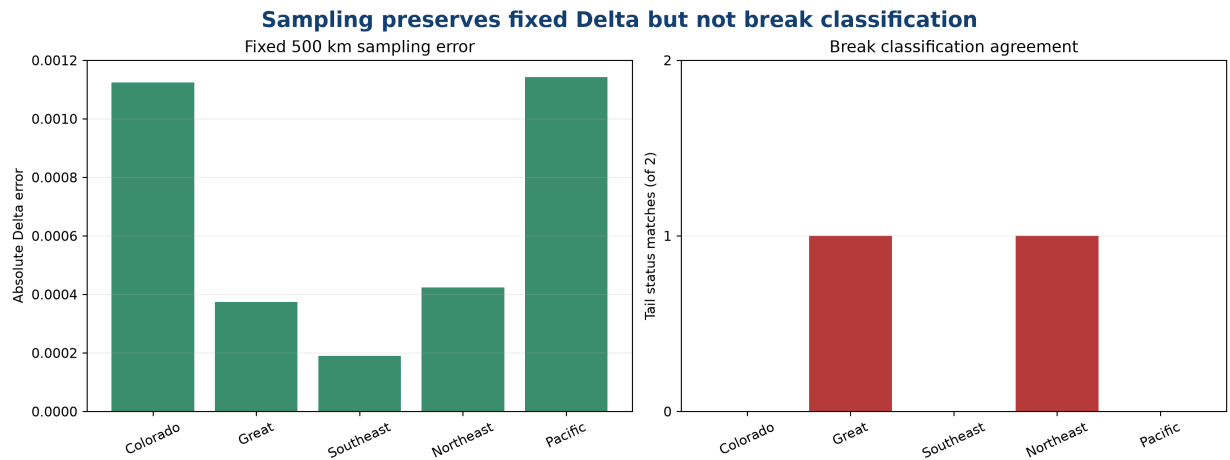
Where is the first reproducible local-to-regional transition?

$$R^*(x, y, t)$$

## 34. Where fixed 500 km was adequate

Sampling reproduced the fixed collapse much better than it reproduced  $R^*$ .

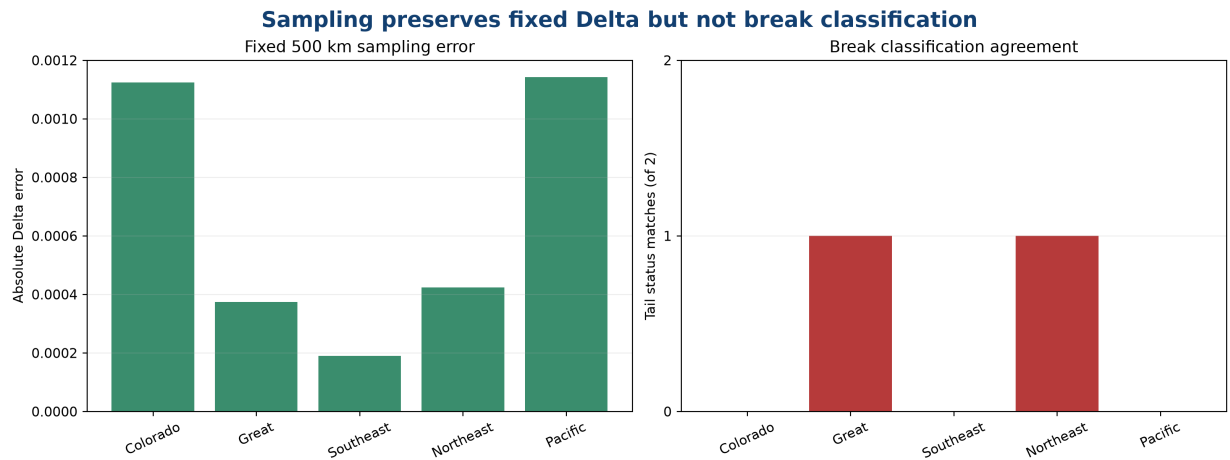
1. Fixed 500 km Delta sampling MAE was 0.000651.
2. All five representative fixed summaries were available.
3. This supports reuse of sampled pairs for fixed summaries, not for break classification.



## 35. Where the fixed radius mattered

The adaptive comparison cannot be generalized yet.

1. Only common-valid sites can answer adaptive-versus-fixed questions.
2. No comparable exhaustive/sampled adaptive Delta sites remained.
3. The scientifically correct result is under-identification, not a forced difference map.



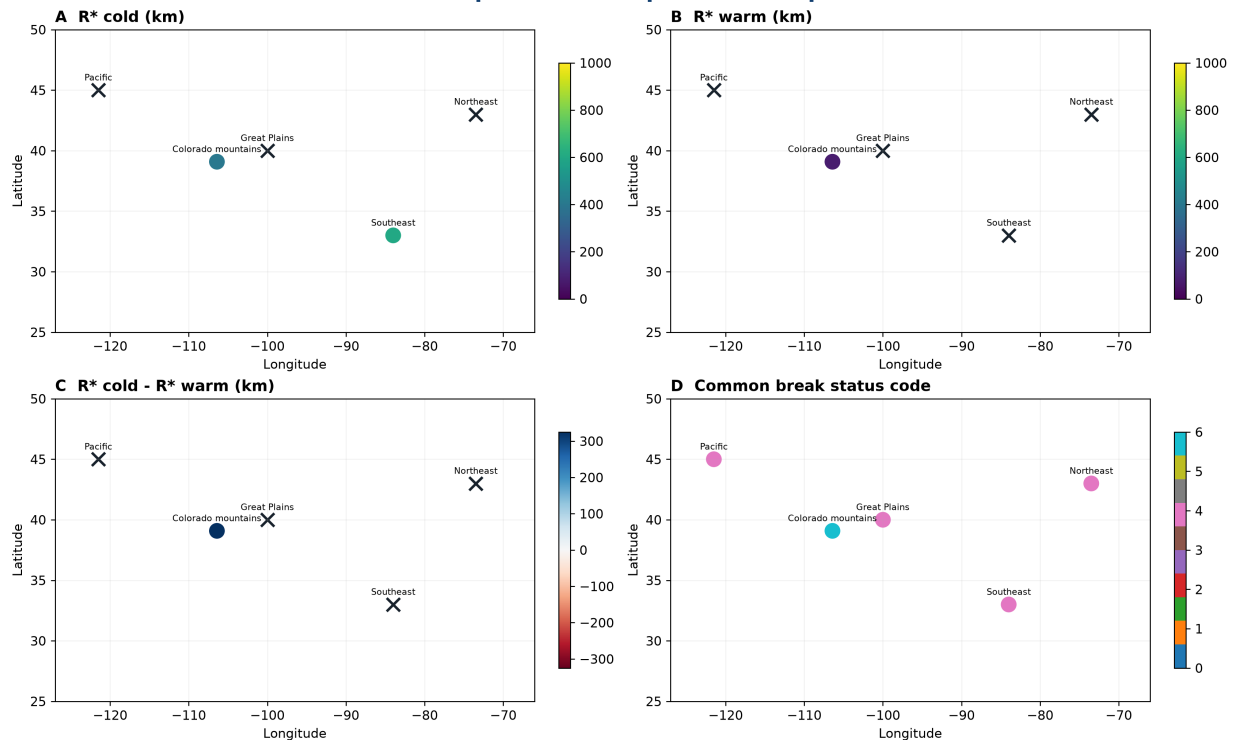
ADAPTIVE COMPARISON UNDER-SUPPORTED

## 36. Spatial variation in characteristic scale

The observed candidates vary, but status instability dominates.

1. Only 30% of tail classifications agreed under sampling.
2. Peak process memory was 3.63 GB during the bounded run.
3. A sampled full-Colorado raster is projected near 550,528,850 pair evaluations.

Scale outputs at five representative pixels

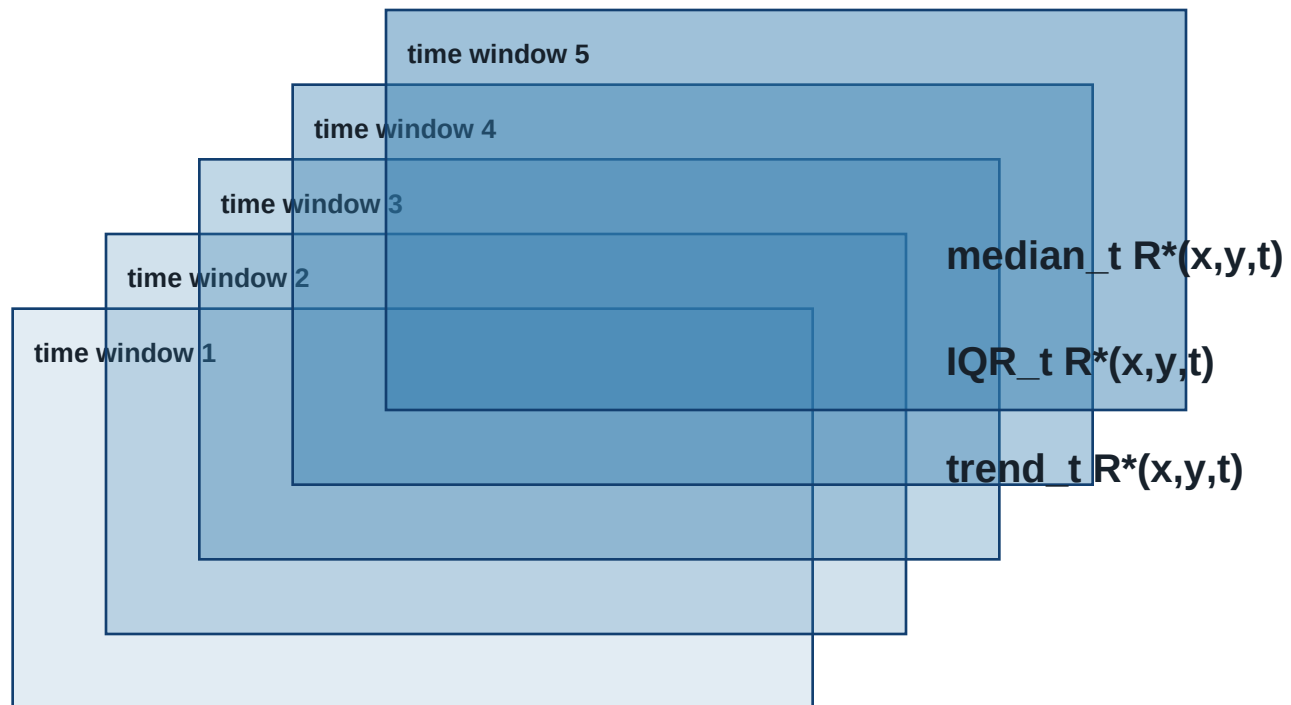


HOLD BEFORE COLORADO OR CONUS SCALE-UP

## 37. Future $R^*(x,y,t)$ extension

The data model already carries a time-window identifier.

1. Repeat the bounded method by independent time windows.
2. Then estimate median, IQR, and trend of  $R^*$  through time.
3. Temporal work should wait until spatial break and sampling gates pass.



NEXT STEP: IMPROVE BREAK IDENTIFIABILITY, NOT SCALE-UP